

Visualization of the Critical Patterns of Missing Values in Classification Data

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Abstract. The patterns of missing values are important for assessing the quality of a classification data set and the validation of classification results. The paper discusses the critical patterns of missing values in a classification data set: missing at random, uneven symmetric missing, and uneven asymmetric missing. It proposes a self-organizing maps (SOM) based cluster analysis method to visualize the patterns of missing values in classification data.

Keywords: Data visualization, missing values, classification, cluster analysis, self-organizing maps, data mining.

1 Introduction

Classification is one of the important techniques of data engineering [6,7]. Commonly, one can rarely find a sample data set for classification that contains complete entries of each observation for all of the variables. The possible reasons of missing values could be numerous, including negligence, deliberate avoidance for privacy, damage, and aversion. The extent of damage of missing data is unknown when it is virtually impossible to return the data source for completion. The incompleteness of data is vital to data quality for classification [9,12]. Indeed, missing data has been an important debatable issue in the classification field [1,2,16,18,19].

There have been several traditional approaches to handling missing values in statistical analysis, including eliminating from the data set those records that have missing values [11] and imputations [4]. However, these traditional approaches can lead to biased classification results and invalid conclusions. Research [17] has pointed out that the patterns of missing values must be taken into account in analyzing incomplete data.

This paper is to examine the patterns of missing values in classification data that are critical for assessing the data quality and validating classification results, and proposes a model to detect and visualize critical patterns of missing values for classification data based on self-organizing maps (SOM) [10]. An experiment

on real-world data is employed to demonstrate the usefulness of the proposed model.

2 Critical Patterns of Missing Values in Classification Data

There are three critical patterns of missing values in classification data. These patterns verify the data quality and influence the validation strategy on classification results.

2.1 Missing at Random (MAR)

Missing values often randomly distributed throughout the sample space. There is no particular assumption on the reason of value missing. Few correlations among the missing values can be observed if the missing values have the MAR pattern. Since values are missing at random, the missing values are not distributed unequally towards a particular class in the classification context. Any imputation analysis would be invalid in this case. The quality of the entire classification data set is determined by the degree of data incompleteness.

2.2 Uneven Symmetric Missing (USM)

Data are often missing in some variables more than in others. Also, missing values in those variables can be correlated. It is difficult to use statistical techniques to detect multi-variable correlations of missing values. An implication of uneven missing for classification is that the data in the subpopulation with those missing data is under representative. In the two-class classification context, uneven missing values may not be biased towards a particular class. In other words, uneven missing values might distribute symmetrically in both classes. The quality of classification data with this pattern of missing values is less homogeneous than that with MAR. Applications of the general classification results based on the complete data set should be cautious for the subpopulation with this pattern.

2.3 Uneven Asymmetric Missing (UAM)

Uneven missing values can be biased towards a particular class. Similar to USM, UAM makes the quality of classification data less homogeneous than that with MAR. More importantly, the general classification results based on the complete data set are highly questionable for the subpopulation with this pattern.

When the data set is high-dimensional, statistical tests for the critical missing patterns might become powerless due to the curse of dimensionality. On the other hand, the interest of a data pattern depends on the purpose of the data analysis and does not solely depend on the estimated statistical strength of the pattern [15].

3 Self-Organizing Maps for Visualizing and Detecting Patterns of Missing Values

3.1 Self-Organizing Maps (SOM)

Cluster analysis, which reveals abnormal attributes of the measurements in the multivariate space, is often used to discover patterns of the data [7]. A definition of clusters for our purposes is: a cluster consists of observations that are close together in terms of Euclidian distance and the clusters themselves are clearly separated [5]. Although there are three general types of statistical clustering methods, namely, hierarchical [8,13], partitioning [8] and overlapping [14], only partitioning is strictly compatible with the definition of clustering in this study. In a typical partitioning clustering method such as the k-means algorithm, an initial specification of cluster “centers” is required. Then observations are assigned to the clusters according to their nearest cluster centers. Cluster “centers” are refined and observations are reallocated. The procedure continues until some type of steady-state is reached. The best known partitioning procedure is the k-means algorithm [8]. A major difficulty in using the k-means algorithm is the selection of parameters for the algorithm a priori. Also, the k-means algorithm acts as a black-box and provides little comparison of the clusters.

The self-organizing maps (SOM) method based on Kohonen neural network [10] has become one of the promising techniques in cluster analysis. SOM-based cluster techniques have advantages over statistical methods in cluster analysis to discover patterns of missing values in classification data. First, classification often deals with very high-dimensional data. The SOM method does not rely on any assumptions of statistical tests, and is considered as an effective method in dealing with high-dimensional data. Secondly, data for classification often do not have regular multivariate distributions, and then the traditional statistical cluster analysis methods have their limitations in these cases. On the other hand, the SOM method has demonstrated its flexibility and usefulness in cluster analysis because of the relaxation of statistical assumptions [10,3]. Thirdly, the SOM method provides a base for the visibility of clusters of high-dimensional data. This feature is not available in any other cluster analysis methods. It is extremely important for our study since we are interested in not only individual clusters, but also the global pattern of clusters such as no-cluster pattern.

SOMs are used to reveal certain useful patterns found in their input data through the unsupervised (competitive) learning process. The SOM method is a typical artificial intelligence technique of cluster analysis. It maps the high-dimensional data onto low-dimensional pictures, and allows human to view the clusters. The neural network depicted in Figure 1 is the two-layer SOM used in this research. The nodes at the lower layer (input nodes) receive inputs presented by the sample data points. The nodes at the upper layer (output nodes) will represent the one-dimensional organization map of the input patterns after the unsupervised learning process. Every low layer node is connected to every upper layer node via a variable connection weight. The lateral interconnections are not included in this study.

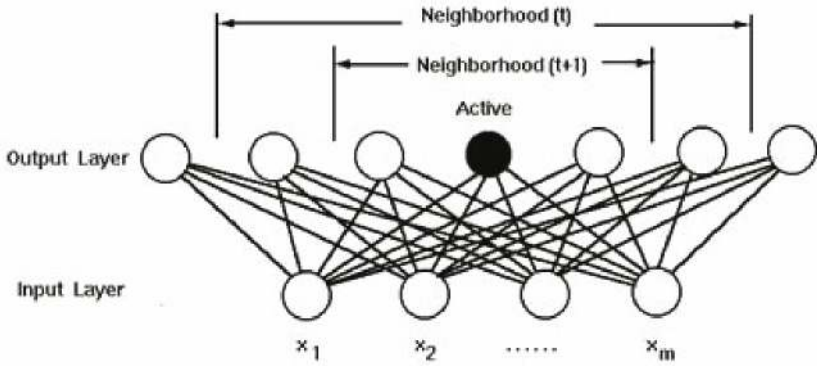


Fig. 1. Two Layer Self-Organizing Maps

The unsupervised learning process in SOM can be briefly described as follows. The connection weights are assigned with small random numbers at the beginning. The incoming input vector presented by a sample data point is received by the input nodes. The input vector is transmitted to the output nodes via the connections. The activation of the output nodes depends upon the input. In a "winner-take-all" competition, the output node with the weights most similar to the input vector becomes active. In the learning stage, the weights are updated following Kohonen learning rule [10]. The weight update only occurs for the active output node and its topological neighbors (Figure 1). In this one-dimensional output case, we assume a linear neighborhood. The neighborhood starts large and slowly decreases in size over time. Because the learning rate is reduced to zero, the learning process will eventually converge. The weights will be organized such that nodes that share a topological resemblance are sensitive to inputs that are similar. The output nodes in SOM will thus be organized and represent the real clusters in the self-organizing map. The reader is referred to [10] for a more detailed discussion.

3.2 Visualizing and Detecting Patterns of Missing Values

The method of visualizing and detecting patterns of missing values is described as follows. The first step of this method is to convert observations with missing values into shadow observations by assigning a value 1 for the data item with missing value and 0 for the data item without missing values. For example, suppose we have an observation with five data items $[2, 4, M, 1, 4]$ where M indicates a missing value. Its shadow observation is $[0, 0, 1, 0, 0]$.

Shadow observations are then presented to the SOM and they are self-organized on the map represented by the output nodes of the SOM. The map is then depicted graphically. The one-dimensional map is almost the same as histogram in statistics. The horizontal axis represents locations of output nodes of the SOM. The height of a bar indicates the number of data points which activate the output node of the SOM at the corresponding location. To determine whether

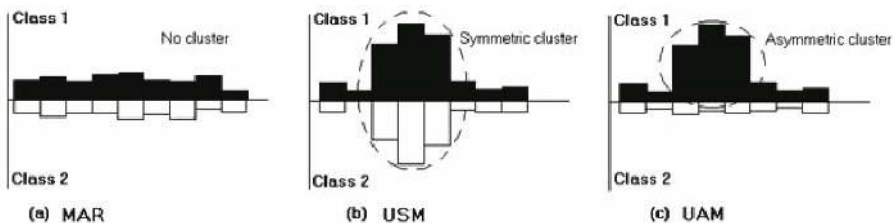


Fig. 2. The Critical Patterns of Missing Values in Classification Data

missing values are biased towards a particular class, we depict the map in two parts, assuming without loss of generality that two classes are involved in the classification of the data set. The upper part represents patterns for one class, and the lower part represents those for the other class.

The three types of critical patterns of missing values in classification data can be illustrated in Figure 2. Figure 2(a) shows a case where no significant cluster is generated by the SOM. Apparently, this pattern shows MAR. The cluster circled in Figure 2(b) represents the pattern USM. It indicates that a substantial number of observations have missing values in certain variables concurrently. The cluster distributes almost evenly in the two spaces corresponding to the two classes, indicating it does not have a significant bias. If one traces these shadow observations back to their original multivariate data, specific variables that are correlated in missing values can be found. The cluster circled in Figure 2(c) represents the pattern UAM since the cluster appears in only one class.

To validate the patterns of missing values visualized from SOM, one might make changes to the parameters of the SOM network, including the neighborhood function and its change rate, and the learning rate and its change rate, to regenerate several self-organizing maps. The conventional split-half validation method can also be used to validate the patterns detected. Using this pattern discovery method, the experiment process and validation are conducted simultaneously, and an adequate result can always be obtained.

4 An Experiment with Real-World Data

Student evaluation of instruction survey methods are widely used at universities to evaluate the teaching performance of instructors. The data used in this experiment came from a student evaluation of instruction survey at a Canadian university. In this case, twenty questions describe the characteristics of an instructor's performance. Each question is rated on a five-point scale for students to answer. A high mark for a question indicates a positive answer to the question. One question of the survey is related to the classification of the instructors into two groups (i.e., effective and ineffective) based on the median value of the entire survey data on this question.

Sample data of 1893 incomplete observations were collected for this experiment. These observations with missing values were converted into shadow

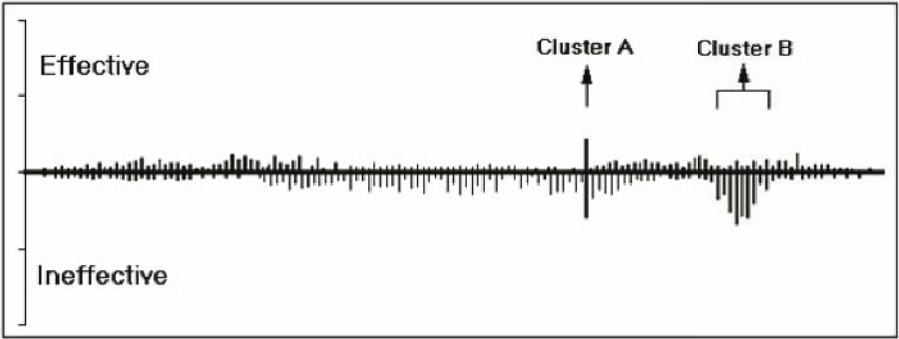


Fig. 3. The Self-Organizing Map for the Missing Values

Table 1. Summary of the Experimental Results

Patterns of Missing Values	Variables Involved	Questions	Interpretations
UAM (Cluster B)	x_{14}	Tests and assignments are reasonable measures of student learning	The four variables were correlated in missing values. The observations with concurrent missing values in these variables were biased towards the ineffective class.
	x_{15}	Where appropriate, student work is graded promptly	
	x_{16}	Where appropriate, helpful comments are provided when student work is graded	
	x_{18}	Tests and assignments provide adequate feedback on student progress	
USM (Cluster A)	x_{20}	The textbook(s) and course material are useful	Missing values in this variable were unbiased. They has little correlation with missing values in other variables

observations for SOM. In making trials of SOM, we started with the extremely large amount of output nodes in accordance with the number of the training samples (i.e., 1893 in this case), and then gradually reduced the number of output nodes to find a clear feature map. Our SOM program was implemented in C++, and had a data conversion interface with Microsoft Excel for graphics visualization. Figure 3 shows the result using the SOM with 200 output nodes, 200 nodes for the initial neighborhood, the initial learning rate of 0.01, and 2000 learning iterations. Since the map presented clear clusters, it was employed for further analysis.

Using the validated cluster analysis result on the SOM, two large clusters were identified, as shown in Figure 3. The original observations corresponding to the

shadow observations of the clusters were extracted, and the nature of missing values in the survey and the hidden correlations of missing values were then identified, as summarized in Table 1. As shown in Table 1, the critical patterns of missing values revealed by the two clusters were UAM and USM. Notably, four variables of the twenty had a correlation of missing values which were biased towards the ineffective class.

5 Conclusions and Discussion

In this paper, we have discussed the critical patterns of missing values in the context of classification. SOM has been considered a useful technique in visualized clustering analysis when statistical clustering methods are difficult to apply. This paper proposes a SOM-based visualized pattern discovery method for missing values in classification data. Through the real-world case, it has been shown the usefulness of this method.

Compared with other traditional methods of treating missing values in classification data, the proposed method has certain advantages. This method provides visual presentations of patterns of missing values so that the user of the method is allowed to detect the quality of data and determine the strategy for validating classification results based on the particular problem domain. It requires no statistical assumptions in comparison with other statistical clustering analysis methods, but can effectively support statistical classification analysis.

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